

BASIC OF CIVIL ENGINEERING (TCE 101)



UNIT- 1 (SYLLABUS)

Introduction and Scope of Civil Engineering:

- Introduction: History of Civil Engineering in India. Civil Engineering Marvels in India and abroad.
- Various Civil Engineering branches and their scope.
- Role of Civil Engineers in the development of the Nation.
- Impact of Infrastructural Development on the economy of our Country.

Ancient India's contribution to Science & Technology

Ancient India was a land of sages, saints and seers as well as a land of scholars and scientists. Ancient India's contribution to science and technology include:

- **Mathematics** - Vedic literature is replete with concepts of zero, the techniques of algebra and algorithm, square root and cube root. Arguably, the origins of Calculus lie in India 300 years before Leibnitz and Newton.
- **Astronomy** - Rig Veda (2000 BC) refers to astronomy.
- **Physics** - Concepts of atom and theory of relativity were explicitly stated by an Indian Philosopher around 600 BC.
- **Chemistry** - Principles of chemistry did not remain abstract but also found expression in distillation of perfumes, aromatic liquids, manufacturing of dyes and pigments, and extraction of sugar.

Ancient India's contribution to Science & Technology

- **Medical science & surgery** - Around 800 BC, first compendium on medicine and surgery was compiled in ancient India.
- **Fine Arts** - Vedas were recited and recitation has to be correct, which gave rise to a finer study of sound and phonetics. The natural corollary were emergence of music and other forms of performing arts.
- **Mechanical & production technology** - Greek historians have testified to smelting of certain metals in India in the 4th century BC.
- **Civil engineering & architecture** - The discovery of urban settlements of Mohenjodaro and Harappa indicate existence of civil engineering & architecture, which blossomed to a highly precise science of civil engineering and architecture and found expression in innumerable monuments of ancient India.
- **Shipbuilding & navigation** - Sanskrit and Pali texts have several references to maritime activity by ancient Indians.

History of Civil Engineering in India

- Indians were familiar with **Sthapaty Kala** i.e., art of construction. The knowledge of civil engineering and architecture acquired by Harappan people is undisputed. The knowledge of construction and its use once again caught momentum in later vedic period and finally culminated into second urban revolution in sixth century B.C.
- The development and growth of the science is best manifested in the form of temples, palaces and forts. Indian architecture and civil engineering reached Baluchistan, Central Asia, Sri Lanka, South-East Asian countries, Korea, Japan etc.
- World heritage sites like Ajanta, Ellora, Khajuraho, Sanchi, Mahabalipuram etc. bear the testimony of excellent planning and execution by architects and craftsmen.
- Water management techniques developed during ancient India indicates the relation between social need and technological progress. Irrigation canals of different sizes, wells of different types (including ring wells), storage tanks etc. were constructed. Sudarsan Lake at Grinar in Kathiwar built in 3rd century B.C. was of enormous size.

History of Civil Engineering in India



Aerial View of Mohenjo-Daro
(4000 years old)

- Square corners (90° corners) !!
- Vertical water flow
- Well developed drainage and sanitary
- Use of standardized bricks (even trapezoidal shape)



2450 BC
Mature Harappan period. 1 well per 3 houses! More than 700 wells in Mohenjodaro. Each house had its own bath and toilet. Population: 40-50,000.

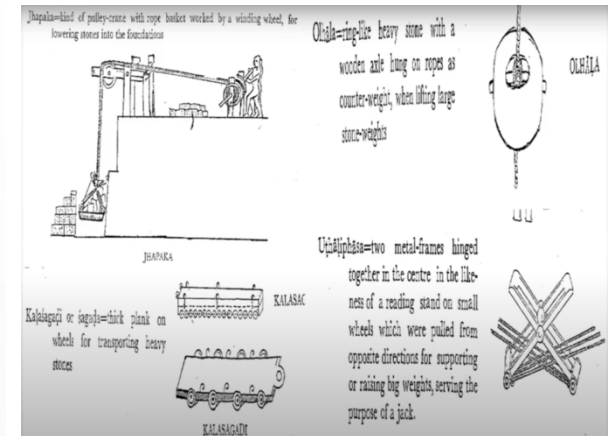
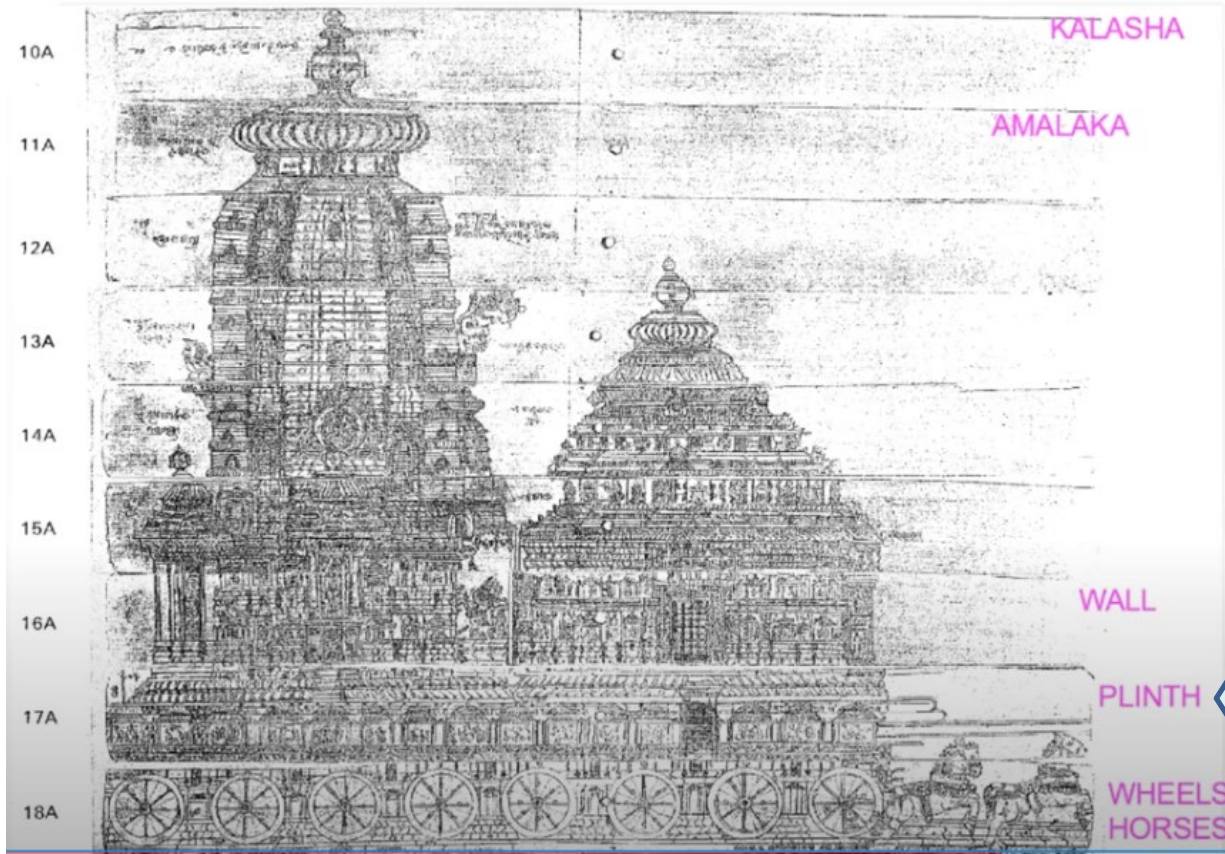


Corbelled culvert for drainage. Surprisingly they did not use arches even though the wells were circular



History of Civil Engineering in India

Ref#2: Introduction to Indian Knowledge System- by Dr. R. N Iyengar (Youtube [Link](#))

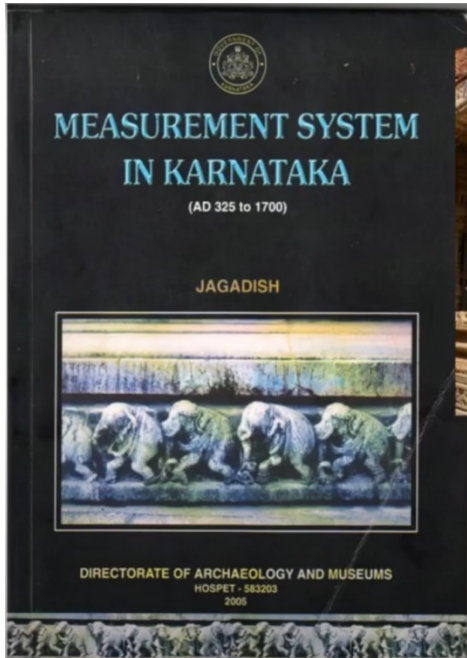


Construction details of
Konark Temple on Palm
leaves

- Details of stones quarry process
- Details of transportation of stone
- Details of canals/ rivers used for transportation
- Details of supervisors, their designations and their signatures
- Details of the payments
- Details of the dates in terms of eclipses
- Details of tools to lift, move, jack-up heavy stones

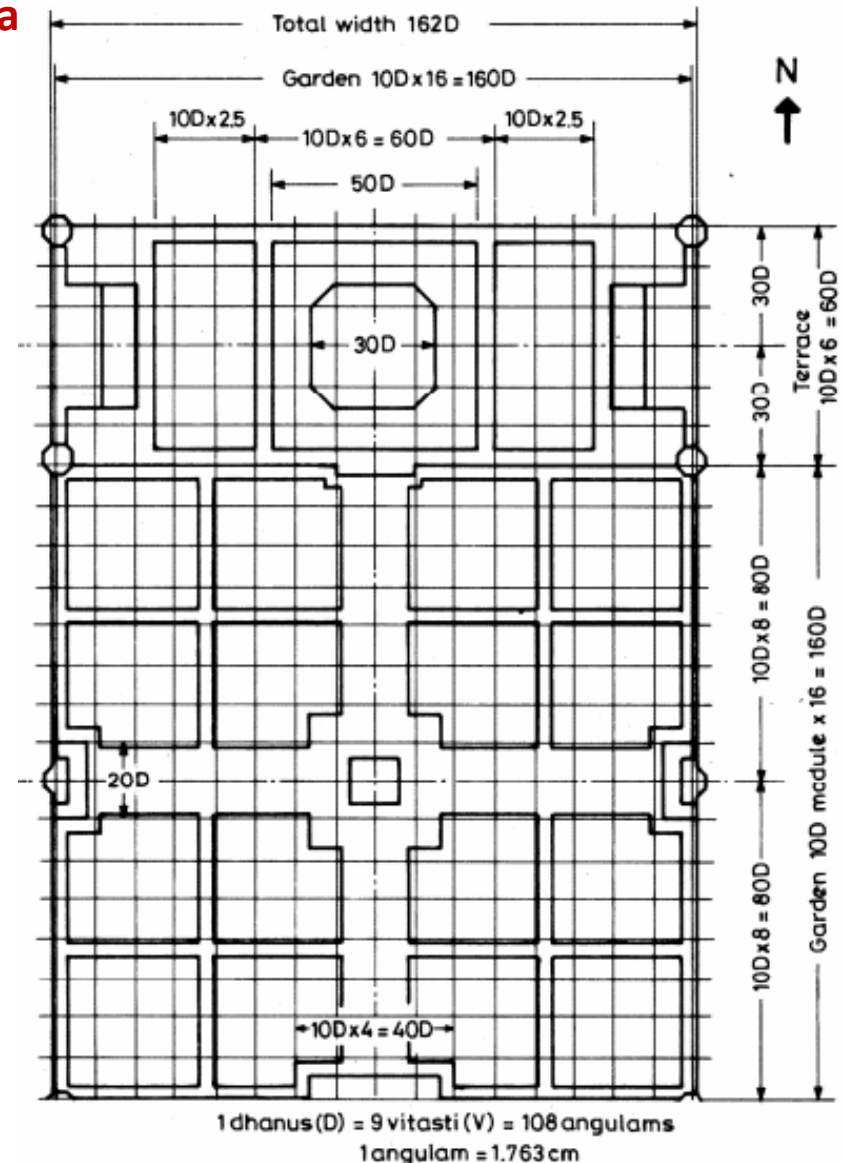
History of Civil Engineering in India

Modular Planning of buildings in Ancient India



EVERY TEMPLE HAS ITS OWN BASIC MEASURE SCALE MARKED & EXHIBITED SOMETIMES WITH NAME AND DETAILS

R. Balasubramaniam, New insights on the modular planning of the Taj Mahal, CURRENT SCIENCE, VOL. 97, NO. 1, 10 JULY 2009



Civil Engineering Marvels in India

- **Bronze Age towns of the Harappan civilization are awe-inspiring,**
- **The bamboo bridges of North-East India,**
- **The step wells of Gujarat and**
- **The palaces of Rajasthan.**
- **Monumental temples built of brick, stone, terracotta, and even chiselled out of hills.**

One cannot help wondering how such large-scale engineering feats were achieved by our ancients. A formidable knowledge system of practice realized in the field and a theory enshrined in large number of Sanskrit texts is an achievement of universal acclaim.

- **town planning,**
- **water management,**
- **Roads and bridges,**

traditional civil engineering provided a sophisticated civilization dimension to Indian heritage that is to be celebrated and emulated by modern engineers.

Civil Engineering Marvels in India

- Bronze Age towns of the Harappan civilization (4000 y ago) are awe-inspiring



Civil Engineering Marvels in India

- **Bamboo bridges of North-East**



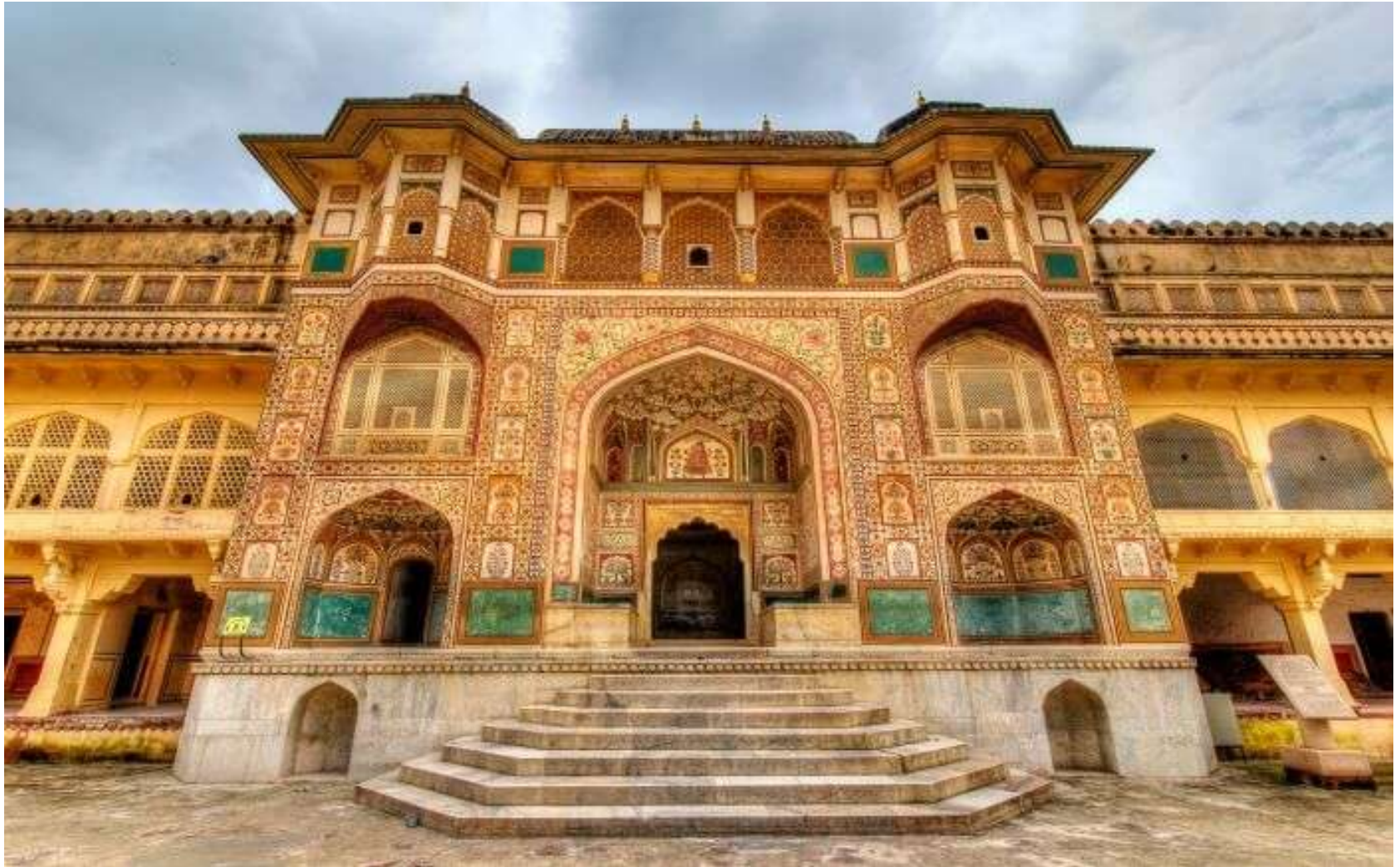
Civil Engineering Marvels in India

- Step wells of Gujarat (600 AD)



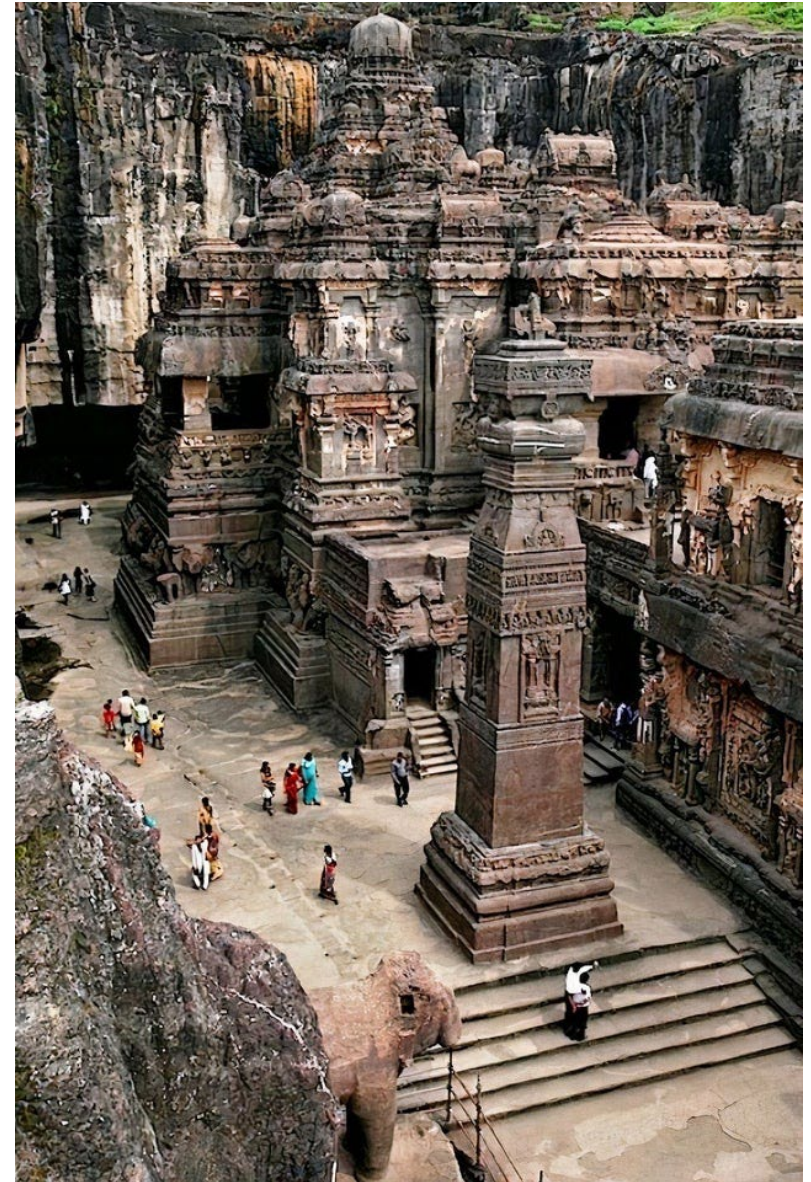
Civil Engineering Marvels in India

- Palaces of Rajasthan



Civil Engineering Marvels in India

- Monumental temples built of brick, stone, terracotta, and even chiselled out of hills



Civil Engineering Marvels in India

Lepakshi Temple, Andhra Pradesh



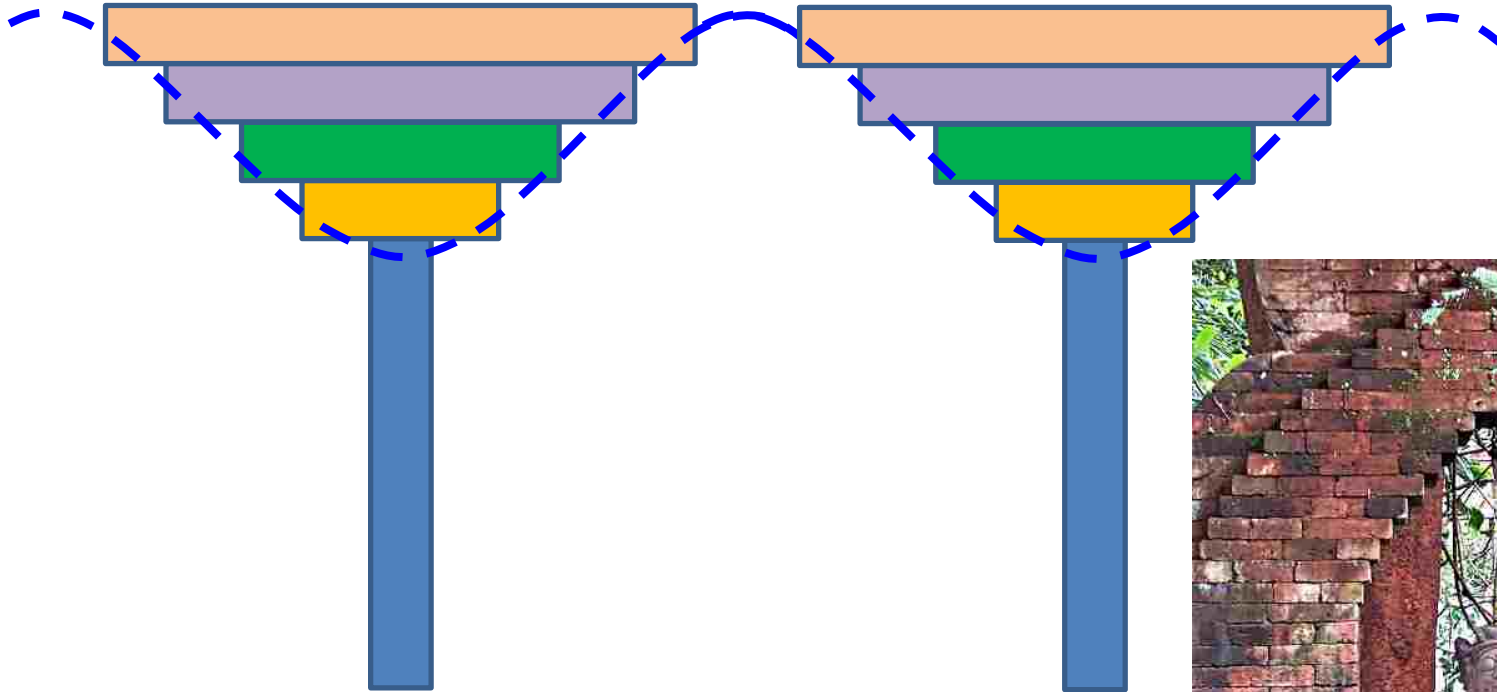
Civil Engineering Marvels in India

Taj Mahal, Agra



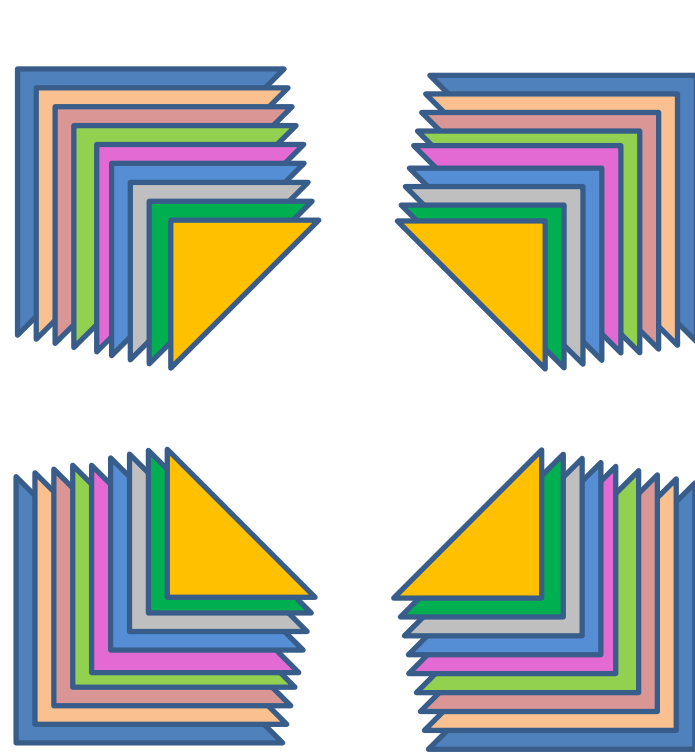
Civil Engineering Marvels in India

- Corbelled Arches

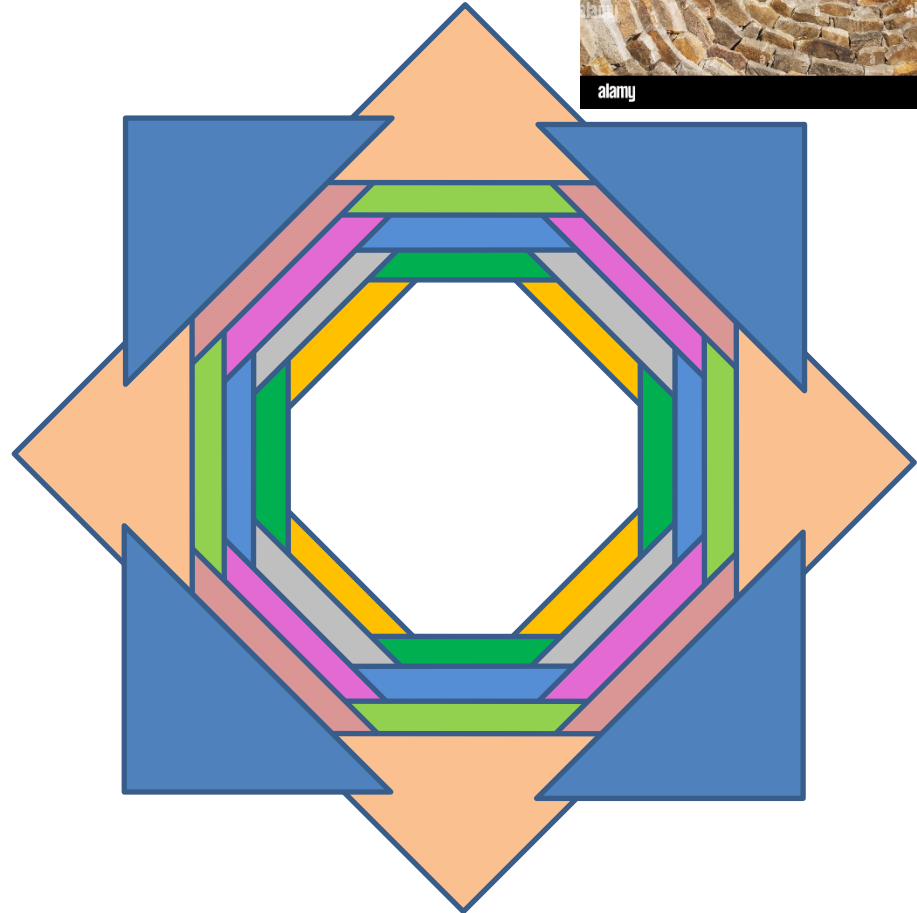


Civil Engineering Marvels in India

Corbelled Domes



Top View



Bottom View

Civil Engineering Marvels in India

- **Town Planning**

Acharya, P. K. (2002).
ManasaraShilpaShāstra. New
Delhi, ND: Munshiram Publishers

The Shilparatna of Srikumara (5/12) mentions about towns such as khetaka or kheta that is situated near rivers or forests by the hills. It has no protective walls. The towns such as *Kharvata* and others have walls that are high and are situated in country side. Arthaśāstra states that it is at the centre of 200 villages. The Viśvakarma VāstuShāstra also deals with other types of towns such as *dronaka* (has king's palace with battlements, four gates, Śiva Temple), *kubja* (fortified towns with king's palace mainly inhabited by merchants), *paṭṭana* (sea port towns), *vāhinīmukha* (located at river mouth, fortified having single gateway with bridge to be crossed) (Shastri & Gadre, 1990, p.117-183).

Shastri, T. G. & Shastri, K. S. (2008). *Shilparatna of Srikumara*.
Trivendrum: SKT series.

Civil Engineering Marvels in India

- Roads and Bridges



Original GT Road passing through Margalla Hills to Kala Chitta Range, Pakistan.

Chandragupta constructed a 2,400 Km long road from Pataliputra (modern Patna) to Takshashila (now in Pakistan). The [Buddhist literature](#) and [Indian epics](#) such as [Mahabharata](#) provide the existence of Grand Trunk road even before the [Maurya Empire](#) and was called [Uttarpatha](#) or the "Northern road". The road connected the eastern region of [India](#) with [Central Asia](#) and [Ancient Greece](#).^[13]- Wikipedia

Civil Engineering Marvels in India

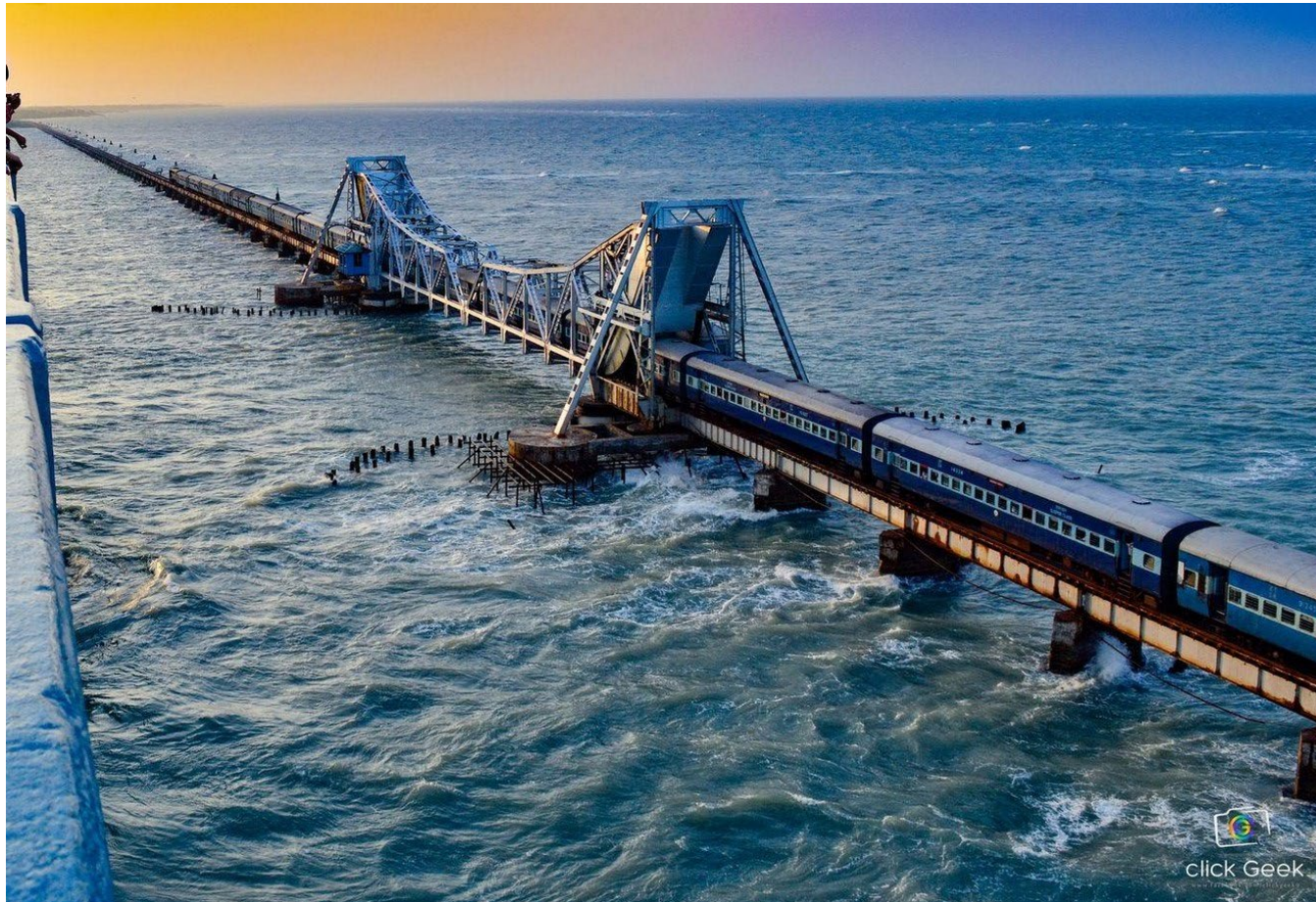
- ATAL Tunnel



Shaped like a horse-shoe, this 9.02-kilometer-long tunnel is the world's longest motorable tunnel. At above 10,000 feet above sea level, this tunnel is named after former Indian Prime Minister, Atal Bihari Vajpayee.

Civil Engineering Marvels in India

Pamban Bridge, Tamil Nadu



Being the second-longest sea bridge of the nation after the Bandra-Worli Sea Link in Mumbai, this bridge connects the island and the mainland..

Civil Engineering Marvels in India

Bandra-Worli Sea Link,



Being the nation's first eight-lane, cable-stayed freeway over the open sea, it took a decade for this bridge to be completed and become fully functional. The height of this stunning sea link is equal to a 43 storied building.

Civil Engineering Marvels in India

Statue Of Unity, Gujarat



Being the tallest statue in the entire world, this structure can be seen within a 7 kilometers radius. There's a museum, a 3-star lodging facility, a food court and a memorial garden on its [premises](#).

Civil Engineering Marvels in India

Bogibeel Bridge, Assam

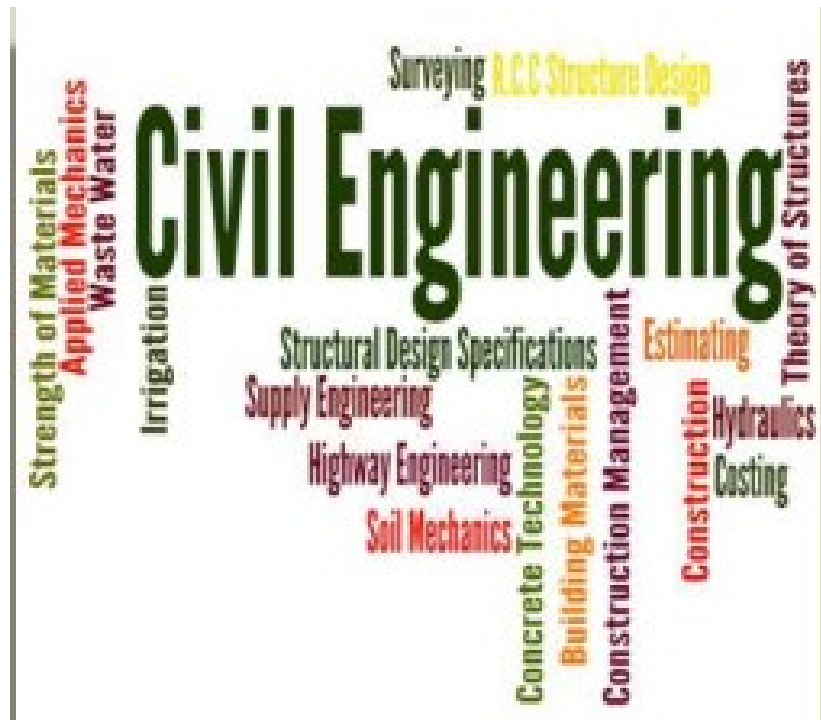


Apart from being the 5th longest bridge in the country, it is also Asia's second-longest rail-cum-road bridge.

Scope of civil engineering

- The main scope of civil engineering or the task of civil engineering is planning, designing, estimating, supervising construction, managing construction, execution, and maintenance of structures like building, roads, bridges, dams, etc.
- Civil engineering is concerned with planning, design and construction for environmental control, development of natural resource, buildings, transportation facilities and other structures required for health, welfare, safety, employment and pleasure of mankind.
- It also includes renovation of the already existing structures. The intention behind this is not only to make the place where we are residing more beautiful for the eye, but also to make the world a much safer place to live in.

Branches of civil engineering



- **SURVEYING & LEVELING**
- **BUILDING, PLANNING AND CONSTRUCTION**
- **STRUCTURAL ENGINEERING**
- **GEOTECHNICAL ENGINEERING**
- **WATER RESOURCES ENGINEERING**
- **TRANSPORTATION ENGINEERING**
- **ENVIRONMENTAL ENGINEERING**

Branches of civil engineering

SURVEYING & GEOMATICS



Branches of civil engineering

SURVEYING AND GEOMATICS

- Geomatics is the new name for an expanded technical area of what used to be called surveying or surveying engineering.
- Surveying, the first step of starting a new civil engineering project, it is a very important branch of civil engineering.
- Surveying is an art which enables the engineer to determine relative positions of points on, above or beneath the earth's surface by means of direct or indirect measurements of distance direction and elevation.
- Surveying is carried out to fix the alignment of road, railway canal. It is also useful in selecting the site for the construction of structures
- Modern surveying instruments like Electronic total station and Geographical Positioning System(GPS) are the modern electronic digital instruments for survey works.
- Remote sensing and Geographical Information system(GIS) are adopted for surveying and planning of many civil engineering projects.

Branches of civil engineering

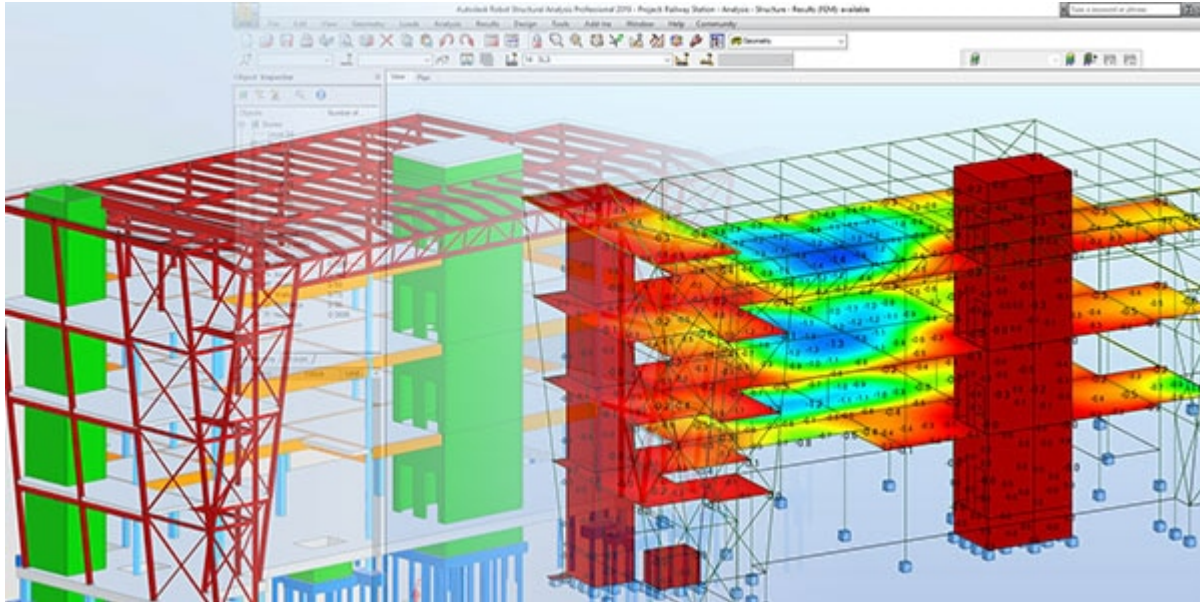
BUILDING PLANNING AND CONSTRUCTION



- Civil engineers are concerned with many types of structures of which buildings are of prime importance.
- Buildings are planned according to the fundamental principles of planning & bylaws of local municipal bodies.
- Building planning requires basic knowledge of principles of architecture.
- Buildings may be residential or public building like school, colleges, government office, hospitals, etc. they are designed according to need of specific occupants and purposes.

Branches of civil engineering

STRUCTURAL ENGINEERING



Branches of civil engineering

STRUCTURAL ENGINEERING

- This Branch of civil engineering deals with structural analysis and design of structures.
- Structural analysis is done to calculate stresses in structural components, on the basis of loads, acting on structures.
- Sections of structural elements like beams, columns, slabs, etc. are designed.
- Structural analysis requires much calculation, hence advanced computing software's are used to carry out structural analysis and design.
- It includes design of reinforced cement concrete (RCC) and steel structures, design of Multistoried buildings, towers, retaining walls, water tanks, bridges requires skills and knowledge of structural engineering.

Branches of civil engineering

GEOTECHNICAL ENGINEERING



Branches of civil engineering

GEOTECHNICAL ENGINEERING

- Geotechnical engineering is that field of civil engineering which deals with soil investigation and design of proper foundations of structures.
- Soil investigation includes collection and testing of soil samples.
- Geotechnical engineering includes measurement of soil parameters and safe bearing capacity. It also includes construction and design of simple foundations, pile foundations, well foundations, caissons, coffer dams, construction of foundation of dams, construction of tunnels, sub base of road, earthen dams, earth related constructions.
- Sound knowledge of geology and geotechnical engineering is necessary for construction of earth related structures.
- As such, the geotechnical engineer is involved in field and laboratory investigations to determine the engineering properties of site soils and other geomaterials and their subsequent use in the analytical study of the problem at hand.

Branches of civil engineering

WATER RESOURCE ENGINEERING



Branches of civil engineering

WATER RESOURCE ENGINEERING

- Water resource engineering means measurements, utilization and development of water resources for agriculture, municipal and power generation purpose.
- It mainly includes irrigation engineering, design of hydraulic structures like dams, canals, etc. Water resource engineering deals with planning designing and developing water resources by constructing several hydraulic structures like dams, barrages, hydropower stations, canal and pipe networks etc.
- It also includes watershed planning, water harvesting techniques, soil conservation and soil reclamation.
- Hydrology is also a part of water resource engineering. Hydrology includes study of sources of water, measurement of rainfall, study of rainfall, runoff, flood control.

Branches of civil engineering

TRANSPORTATION ENGINEERING



Branches of civil engineering

TRANSPORTATION ENGINEERING

- Transportation means movement of passengers and goods by means of vehicles on land, ship on water and aircrafts in air.
- Transportation Engineering is that branch of Civil engineering which deals with planning, designing and construction of roads, bridges, railways, tunnels, harbours, ports, docks, runways, and airports.
- As for development of any nation good transportation network is of prime importance. Study of various construction materials used in construction of roads, traffic engineering are also considered under transportation Engineering.

Branches of civil engineering

ENVIRONMENTAL ENGINEERING



Branches of civil engineering

ENVIRONMENTAL ENGINEERING

- Environmental Engineering deals with pollution control and public health engineering.
- Different types of pollutions are water, air, noise and other pollution.
- Due to large scale industrialization, population growth, rapid urbanization and several other human activities like construction, mining, transportation, environment gets polluted.
- Environmental engineering deals with technologies & facilities which are engaged in reducing pollution.
- Environmental engineering includes design, construction and maintenance of water treatment plant, waste water treatment plant, water distribution network and sewerage system, it also deals with solid waste management in towns and cities.
- Public health engineering includes water treatment, water distribution network, & solid waste management.

Role of Civil Engineers in the development of the Nation

If infrastructure forms the backbone of any nation, then Civil Engineers can be truly termed as Nation Builders.

- Building roads, making new flyovers, constructing dams are some of important tasks undertaken by Civil Engineers which can be linked directly to progress of any nation at global arena.
- With the changing times and new technologies being introduced in the field of construction, the role of Civil Engineers has evolved a lot.
- Today's Civil Engineers are not only confined to supervisor's role but also lays blueprint by using the modern technology & techniques to make impossible project a reality.

Delhi Metro is best example of extreme engineering and is considered to be one of the finest marvels constructed by Civil Engineers of India. Who would have imagined that a Mass Public Transportation System could be constructed in the unplanned locations of Delhi, where even it is difficult to take a car from one location to another? But with the use of technology and hard thinking of Civil Engineers, Delhi Metro is running successfully and the project has truly turned out to be Pride of India.

Infrastructure for Nation Development

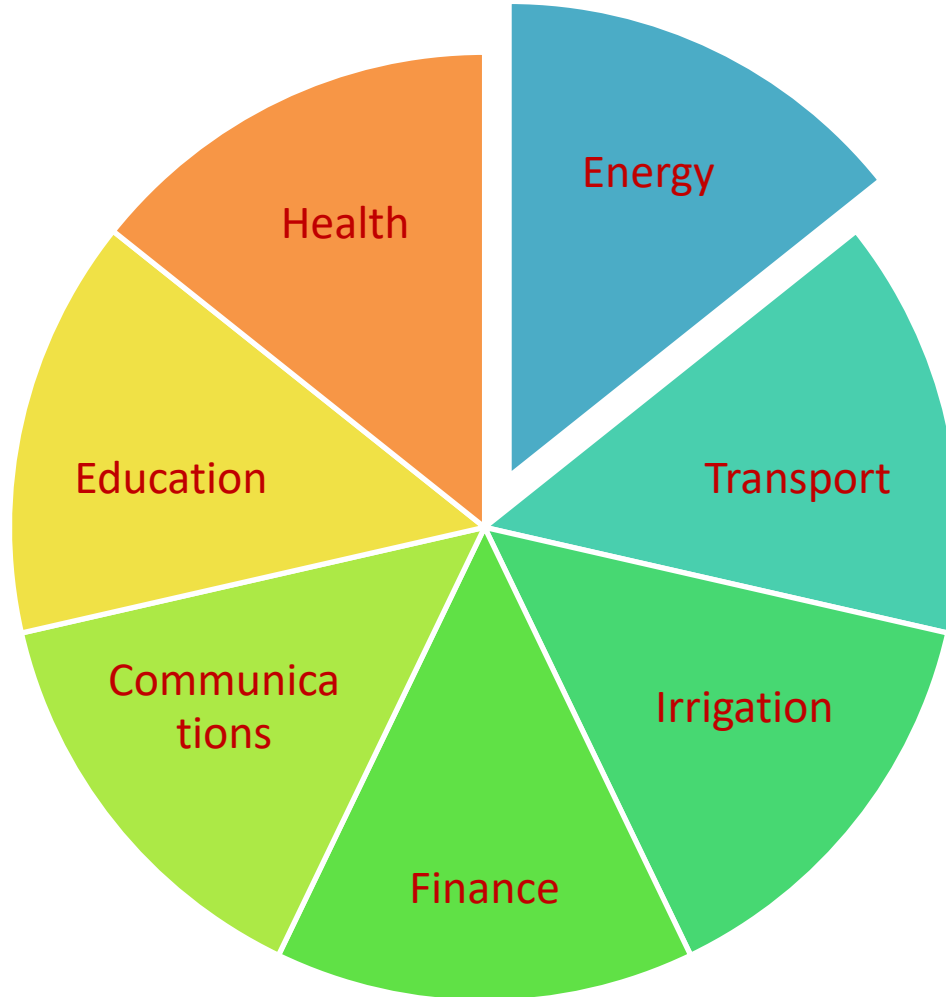
Civil engineers play a major role in the infrastructures development of a country. All structures constructed in the past exhibit the path of civilization and current infrastructures development express the practices followed by civil engineers.

Infrastructure can be defined as activities that provide society with services necessary to conduct daily life and to engage in productive activity and development in a country's economy.

ECONOMY ↔ INFRASTRUCTURE

Infrastructure for Nation Development

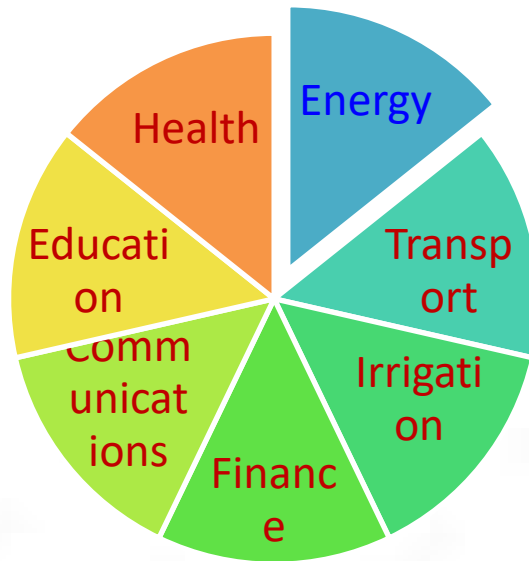
ECONOMY ↔ INFRASTRUCTURE



The knowledge of basic areas of civil engineering can be of great use in providing the infrastructural facilities where constructional aspects are involved for development of regions.

Infrastructure for Nation Development

ECONOMY ↔ INFRASTRUCTURE



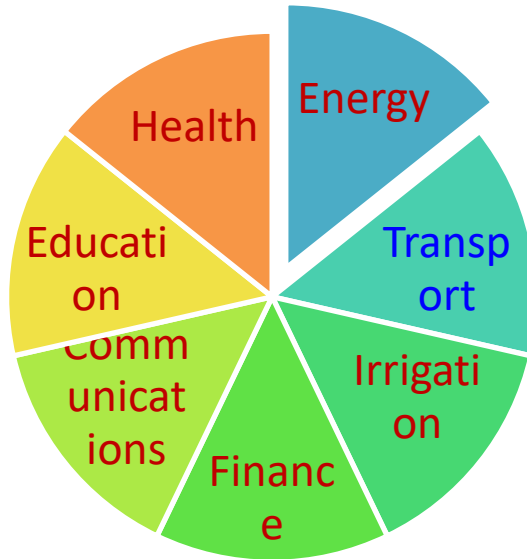
- Electrical energy is so far best for transmission of power. Even chemical energy of coal/petrol/ diesel is converted into electrical for long distance transmission .
- The construction of dams (for hydro-electric power), wind turbines (for wind energy), nuclear reactors (for nuclear energy), solar power plants, Thermal power plants (for coal/diesel based) requires Civil Engineering

Also the power produced from all these sources are to be transmitted and distributed to the user through transmission line towers and distribution poles



Infrastructure for Nation Development

ECONOMY ↔ INFRASTRUCTURE

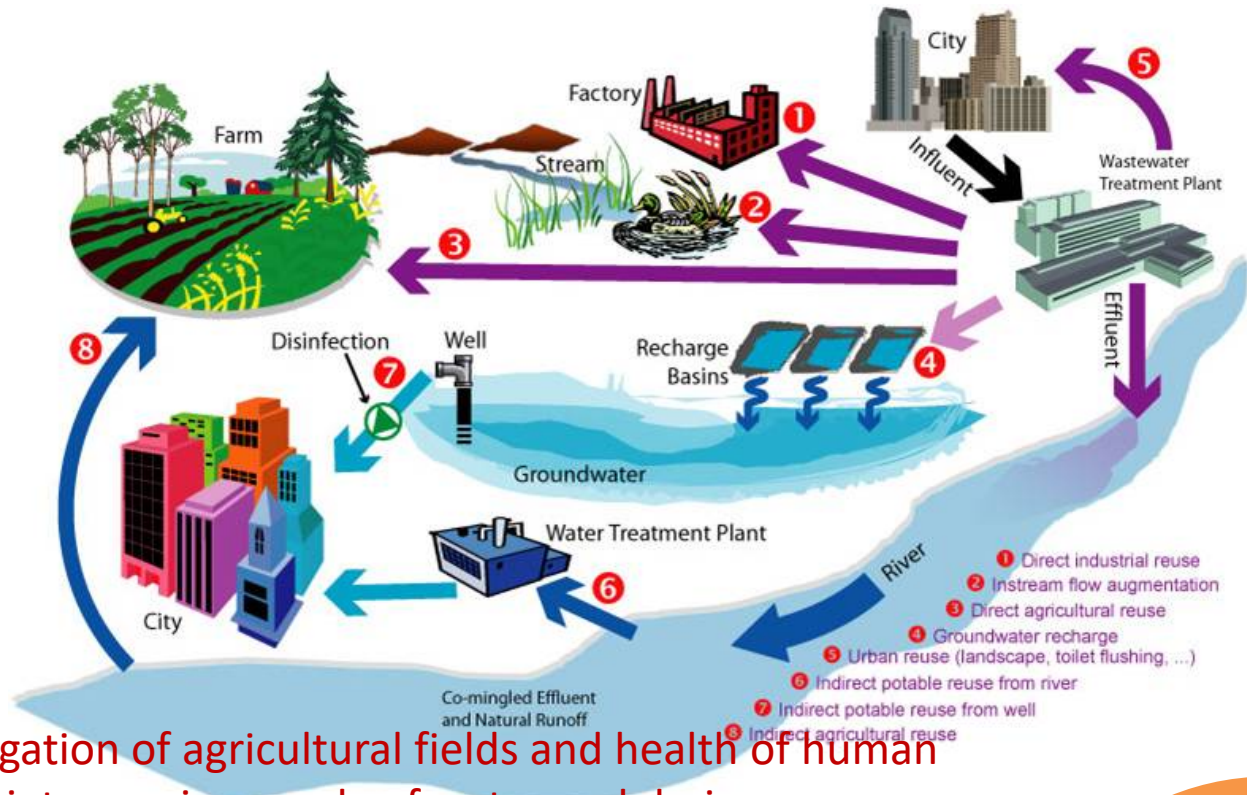


- Transportation is a specialization in Civil Engineering
 - Roadways
 - Railways
 - Inland navigation
 - Marine transportation
 - Airways



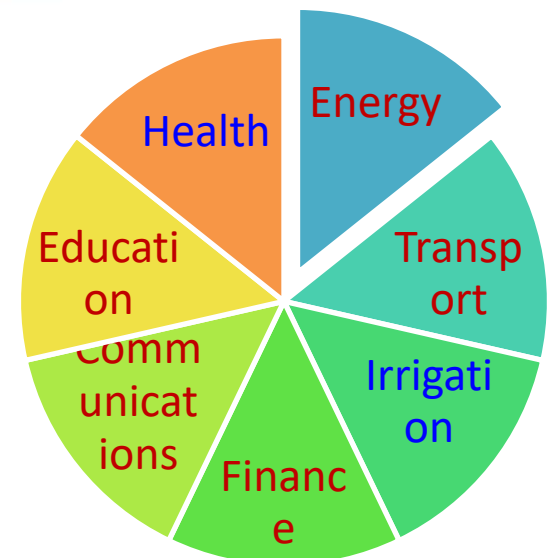
Infrastructure for Nation Development

ECONOMY ↔ INFRASTRUCTURE



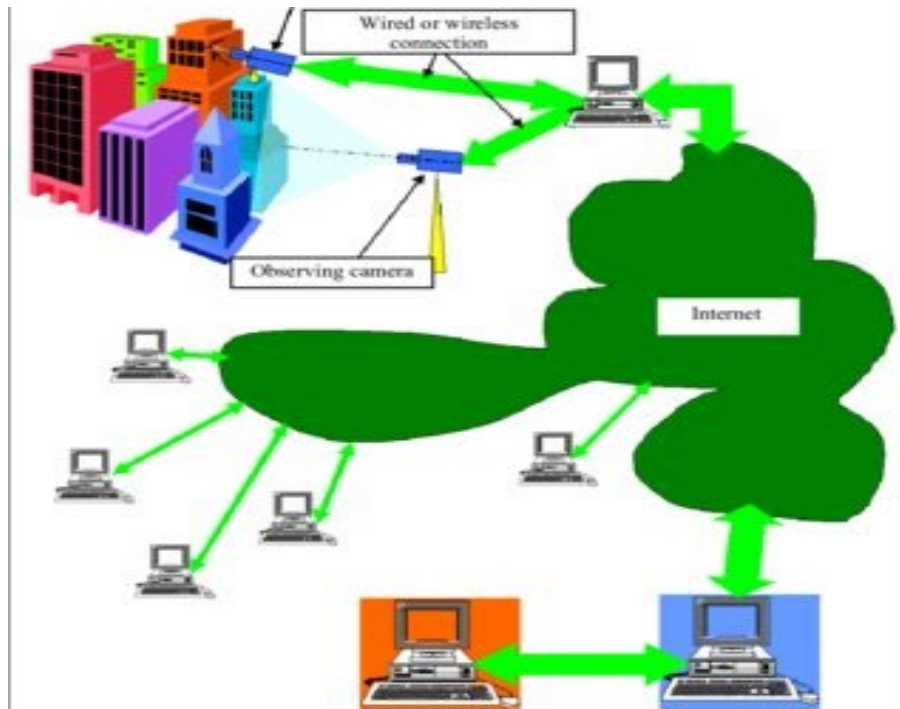
- Irrigation of agricultural fields and health of human society require supply of water and drainage system

- Construction of dams/ reservoirs
- Canals for distributing the water
- Cross-drainage works
- Water treatment plants
- Pipe network for Water supply
- Sewage disposal and treatment
- Maintenance of River banks for flood control



Infrastructure for Nation Development

ECONOMY ↔ INFRASTRUCTURE



- Communication systems and Civil Engineering
 - Wired communication
 - Fiber optic network
 - Wireless Communication
 - (microwave towers)
 - Satellite launching stations



Infrastructure for Nation Development

- Good **surface communication** links such as tar or concrete roads.
- Provision of **water supply distribution** system i.e. construction of water storage reservation or sumps, laying of underground pipes etc.
- Provision of a **drainage system** which may include construction of surface drains as subsurface drains for the disposal of wastewater.
- Supply of **electrical power** for which construction of transmission line towers, construction of electrical substations.
- Providing **inland communications lines**, i.e telephone lines etc.
- Construction of **recreational places** e.g gardens, parks etc.

ECONOMY ↔ INFRASTRUCTURE

Types and classification of Structures

- Buildings
- Towers
- Chimneys
- Bridges
- Dams
- Retaining Walls
- Water Tanks
- Silos, Stakes
- Cable/Rope Ways
- Tunnels
- Roads
- Railways
- Runways
- Offshore Structures
- Naval Structures
- Pipelines
- Inland Waterways.

Types and classification of Structures

- Buildings: To house people and their things
- Towers: To support power transmission lines
- Chimneys: To transfer hot gases to a higher altitude
- Bridges: To provide access over an obstruction
- Dams: To retain water in a valley
- Retaining Walls: To retain soil from falling on..
- Water Tanks: To retain water for storage
- Silos: To store food grains, cement, etc
- Cable/Rope Ways: To provide access between hills
- Tunnels: To provide access through a hill
- Roads: To provide support to moving vehicles
- Railways: To provide support to moving trains

Types and classification of Structures

- Runways: To support aircraft during takeoff/landing
- Offshore Structures: To support machinery over sea
- Naval Structures: To support and anchor ships
- Pipelines: To provide channel for flow of liquids
- Inland Waterway: To provide channel for ships

Types and classification of Structures

- Buildings: To house people and their things



- Should be stiff and strong
- Should be safe under EQ, rain, wind, heat& cold
- Should be well ventilated
- Should have good lighting
- Should support water supply facilities

Types and classification of Structures

- Towers: To support power transmission lines

At sufficient height. Also for mobile communication from high elevations



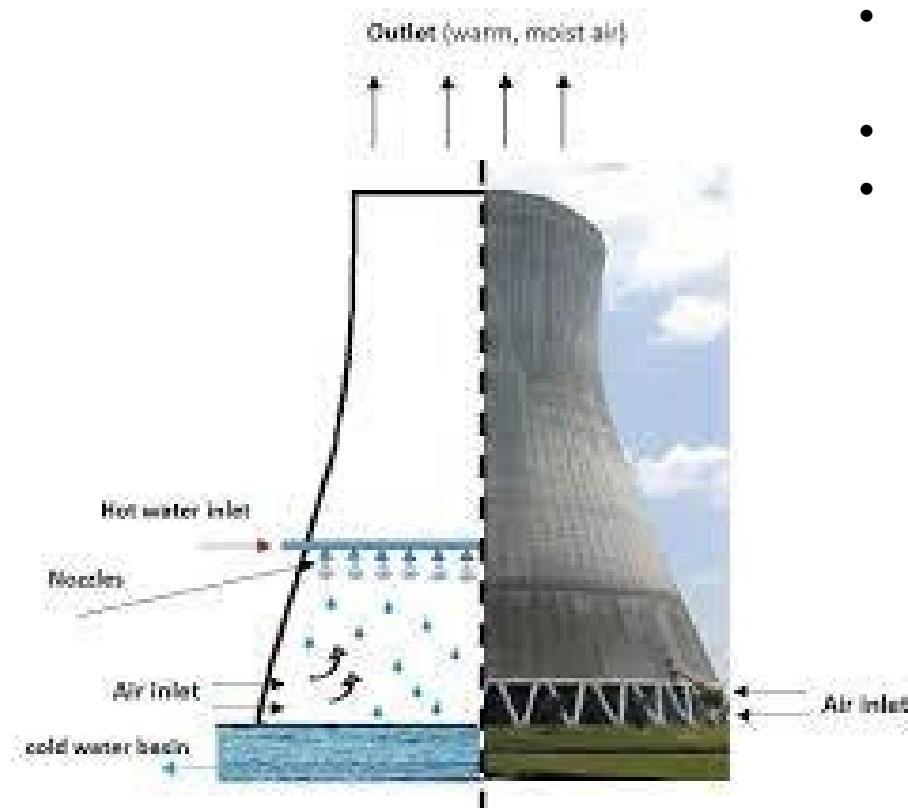
- Should be high enough for safety to men and animals
- Should be stiff and strong
- Should be able to resist accident loads
- Should not collapse even if adjacent towers collapse

Types and classification of Structures

- Chimneys: To transfer hot gases to high altitude

At sufficient height.

- Should be high enough so that the gases do not come back
- Should be stiff and strong to stand under winds
- Should be able to support the water used for cooling



Types and classification of Structures

Home work: Prepare such slides
for rest of the structures